

Question #01

Down Counter D-FF

Present State	Next State $D_3 D_2 D_1 D_0$	D.S $Q_3 Q_2 Q_1 Q_0$	N.S $D_3 D_2 D_1 D_0$
1 1 1 1	1 1 1 0	0 1 1 1	0 1 1 0
1 1 1 0	1 1 0 1	0 1 1 0	0 1 0 1
1 1 0 1	1 1 0 0	0 1 0 1	0 1 0 0
1 1 0 0	1 0 1 1	0 1 0 0	0 0 1 1
1 0 1 1	1 0 1 0	0 0 1 1	0 0 1 0
1 0 1 0	1 0 0 1	0 0 1 0	0 0 0 1
1 0 0 1	1 0 0 0	0 0 0 1	0 0 0 0
1 0 0 0	0 1 1 1	0 0 0 0	1 1 1 1

D_3

$Q_3 Q_2$	00	01	11	10
00	1			
01				
11	1	1	1	1
10		1	1	1

D_2

$Q_3 Q_2$	00	01	11	10
00	1			
01		1	1	1
11		1	1	1
10	1			

D_1

$Q_3 Q_2$	00	01	11	10
00	1		1	
01	1		1	
11	1		1	
10	1		1	

$$D_3 = Q_0 Q_3 + Q_1 Q_2 + Q_2 Q_3 + \overline{Q_0} \overline{Q_1} \overline{Q_2} \overline{Q_3}$$

$$\Rightarrow Q_3 (Q_0 + Q_1 + Q_2) + \overline{Q_3} (\overline{Q_0 + Q_1 + Q_2})$$

1 SOP $\rightarrow$ POS

$$ABC \rightarrow \overline{A} + \overline{B} + \overline{C}$$

$$1 AB + \overline{A} \overline{B} = A \text{ XOR } B$$

$\Rightarrow$

$$\Rightarrow Q_3 \oplus (Q_0 + Q_1 + Q_2)$$

$$D_2 = Q_0 Q_2 + Q_1 Q_2 + \overline{Q_0} \overline{Q_1} \overline{Q_2}$$

1 SOP $\rightarrow$ POS

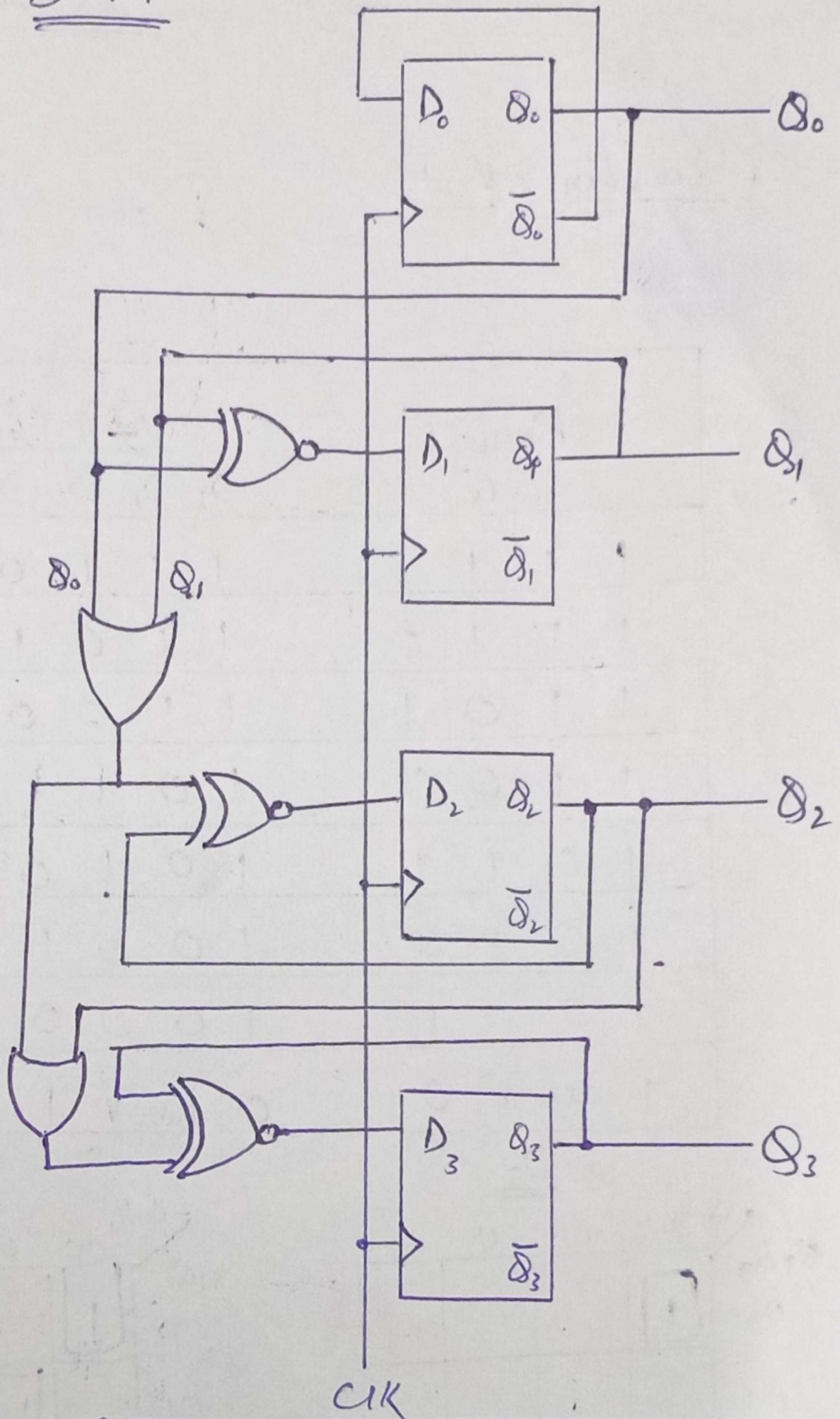
$$\Rightarrow Q_2 (Q_0 + Q_1) + \overline{Q_2} (\overline{Q_0 + Q_1})$$

$$\Rightarrow Q_2 \oplus (Q_0 + Q_1)$$

$$D_1 = Q_0 Q_1 + \overline{Q_0} \overline{Q_1}$$

$$D_1 = Q_0 \oplus Q_1$$

Using D-FF



Question #1

P.S

$$\begin{matrix} Q_3 & Q_2 & Q_1 & Q_0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \end{matrix}$$

N.S

$$\begin{matrix} D_3 & D_2 & D_1 & D_0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{matrix}$$

P.S

$$\begin{matrix} Q_3 & Q_2 & Q_1 & Q_0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{matrix}$$

N.S

$$\begin{matrix} D_3 & D_2 & D_1 & D_0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \\ \times & \times & \times & \times \end{matrix}$$

					$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
Γ_3	Γ_2	Γ_1	Γ_0		00					
0	0	0	1		01			1		
0	0	0	1		11	X	X	X	X	
0	1	1	1		10		1	X	X	
0	0	0	1							
0	0	0	1							
1	1	1	1							
0	0	0	1							
1	0	0	1							

Γ_3
 $\Gamma_2 = Q_0 Q_3 + Q_0 Q_1 Q_2$

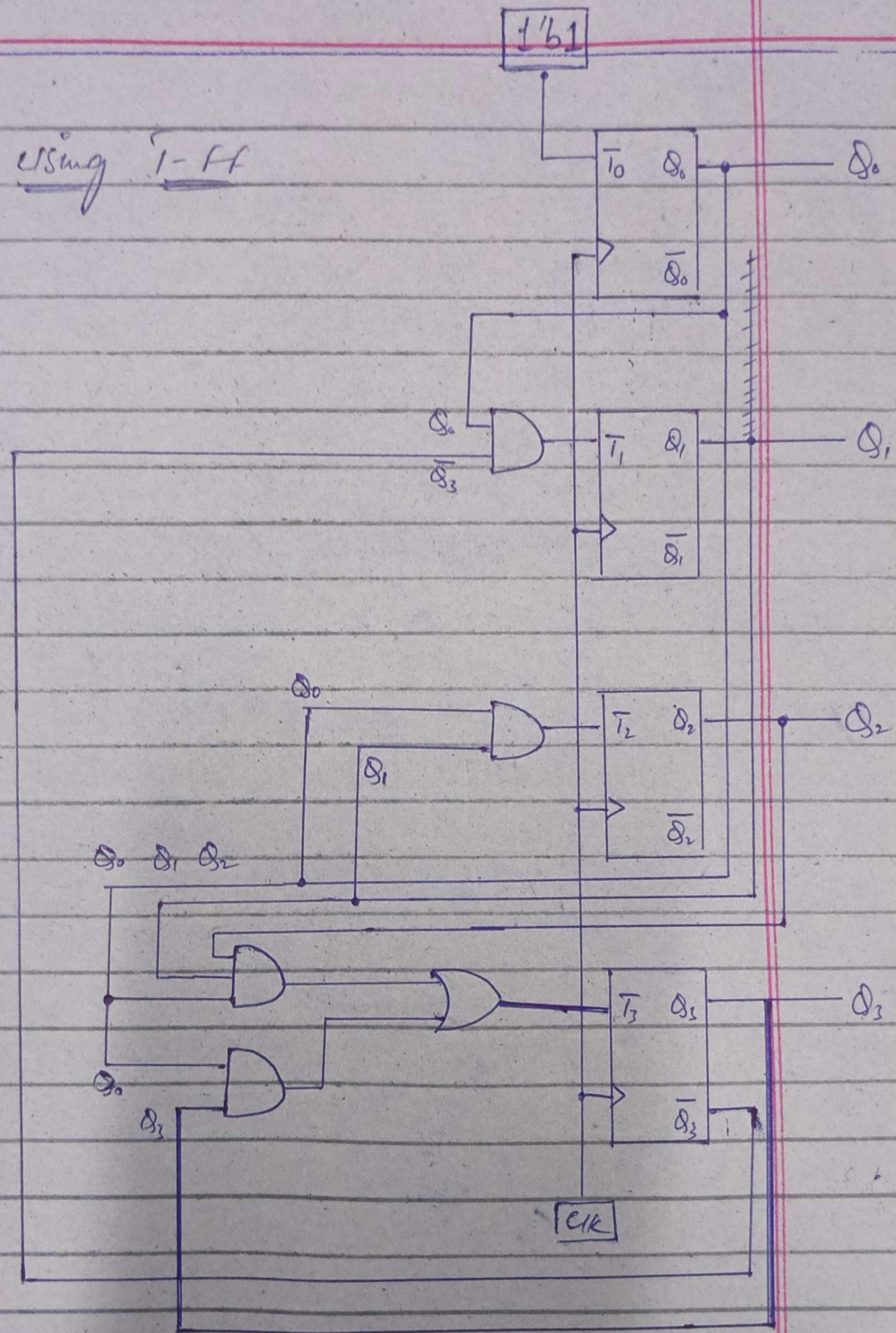
Γ_2
 $\Gamma_2 = Q_0 Q_1$

					$Q_3 Q_2$	$Q_1 Q_0$	00	01	11	10
00							1	1		
01							1	1		
11	X	X	X	X						
10									X	X

$\Gamma_0 = 1$

$\Gamma_1 = \overline{Q_3} Q_0$

using 1-FF



Question #05

2's Complement -

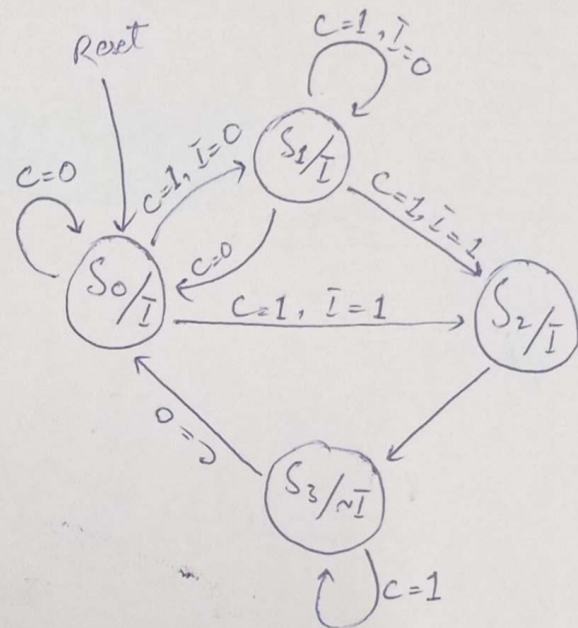
Let say we have an 8-bit number 10011100 and we need to find out the 2's complement.

10011100
01100011 $\rightarrow$ 1's Complement

+1
01100100 $\rightarrow$ 2's Complement

So the basic logic is when first one (1) is detected then the remaining all bits are inverted.

$Q_1, Q_0, C, \bar{I}$	D_1, D_0	X
0 0 0 0	0 0	0
0 0 0 1	0 0	1
0 0 1 0	0 1	0
0 0 1 1	1 0	1
0 1 0 0	0 0	0
0 1 0 1	0 0	1
0 1 1 0	0 1	0
0 1 1 1	1 0	1
1 0 0 0	x x	x
1 0 0 1	x x	x
1 0 1 0	x x	x
1 0 1 1	x x	x
1 1 0 0	0 0	1
1 1 0 1	0 0	0
1 1 1 0	1 1	1
1 1 1 1	1 1	0



Q_1, Q_0	$C, \bar{I}$	00	01	11	10
00					
01					
11					
10					

Q_1, Q_0	$C, \bar{I}$	00	01	11	10
00					
01					
11					
10					

Q_1, Q_0	$C, \bar{I}$	00	01	11	10
00					
01					
11					
10					

$$Q_1, \bar{I}$$

$$\bar{Q}_1, \bar{I} + Q_1, \bar{I}$$

$$D_1 = Q_1, C + C, \bar{I}$$

$$D_1 = C(I + Q_1)$$

$$D_0 = Q_1, C + C, \bar{I}$$

$$D_0 = C(I + Q_1)$$

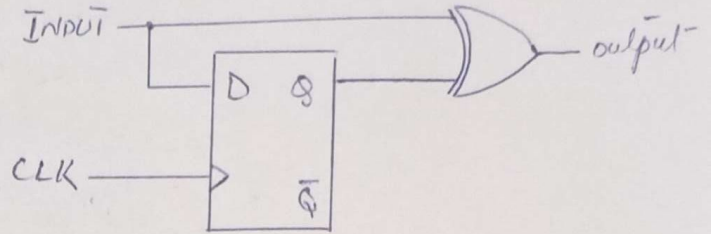
$$X = \bar{I} \oplus Q_1$$

Questions #04

In the waveform we observe that if input changes the output is high. Means it is the XOR of present and current state. So we use D-FF for this problem.

Input	<u>P.O.S</u>	<u>N.O.S</u>	
D	Q_n	Q_{n+1}	Output
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

Truth Table of D-FF



δ_n	δ_{n+1}	Δ
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Table

Q D-FF

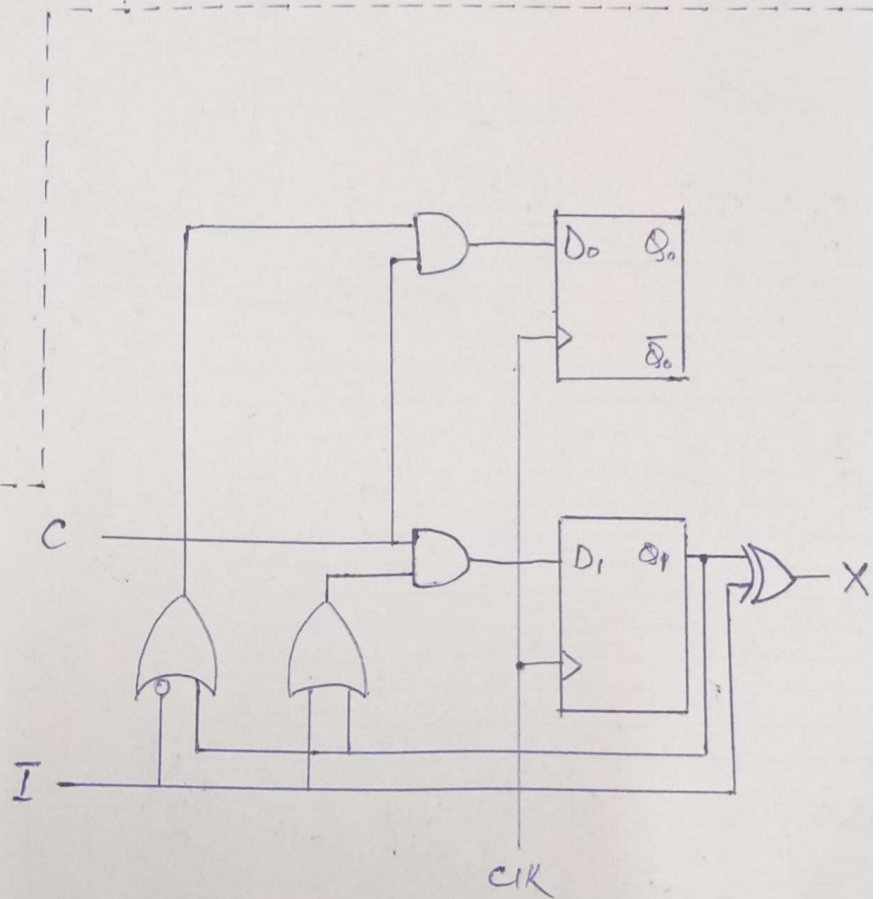


Diagram of Question #05